How-To Library

Informatica

Metadata Access Service Quick Reference

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Abstract

Informatica Big Data Management® provides access to the Hadoop environment to perform activities such as data integration. Big Data Management makes use of various application services to access and integrate data from the Hadoop environment at design time and at run time. This article explains Metadata Access Service used to access and preview metadata from the Hadoop environment at design time.

Supported Versions

- Informatica Big Data Management 10.2.1

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Metadata Access Service Overview

The Metadata Access Service is a user-managed service that allows the Developer tool to access Hadoop connection information to import and preview metadata.

The Metadata Access Service contains information about the Service Principal Name (SPN) and keytab information if the Hadoop cluster uses Kerberos authentication. You can create one or more Metadata Access Services on a node. Based on your license, the Metadata Access Service can be highly available.

The following Informatica products use the Metadata Access Service to extract metadata at design time:

- Big Data Management
  - PowerExchange for HBase
  - PowerExchange for HDFS
  - PowerExchange for Hive
  - PowerExchange for MapR-DB
- Big Data Parser
- Big Data Quality

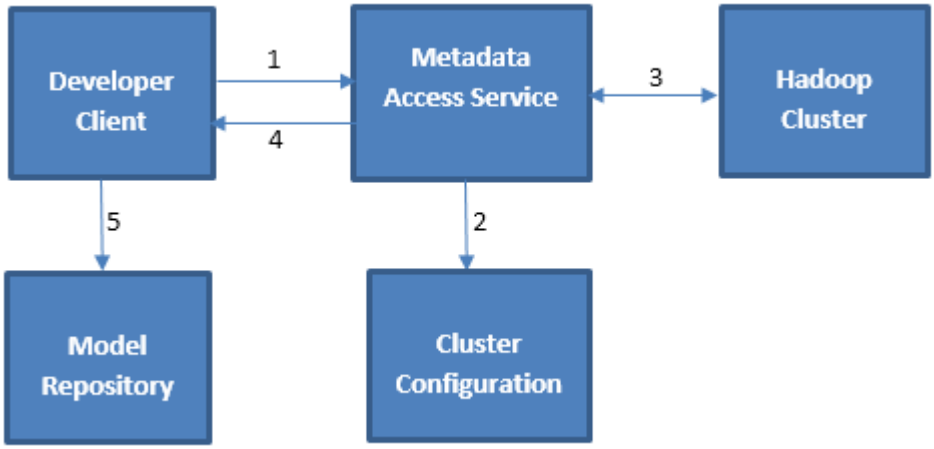
- Big Data Streaming
  - MapRStreams Connection

Create a Metadata Access Service before you import HBase, HDFS, Hive, MapR-DB, and MapRStreams objects from a Hadoop distribution. The Metadata Access Service is not used at run time. When you run HBase, HDFS, Hive, or, MapRStreams mappings, the Data Integration Service communicates with the Hadoop distribution and performs the data integration task.

Metadata Access Service Process

The Metadata Access Service receives requests from the Developer tool at design time to fetch an object metadata from a Hadoop cluster. The Metadata Access Service access the Hadoop cluster and provides the object metadata to the Developer tool.

The Metadata Access Service uses HTTP or HTTPS to communicate with Hadoop clusters and the Developer tool that sends metadata access requests. The following image shows how the Metadata Access Service components complete job requests:



1. When importing a data object, the Developer tool sends a request to the Metadata Access Service along with the reference to the connection object.
2. The Metadata Access Service accesses the cluster configuration defined in the connection object.
3. The Metadata Access Service uses the Hadoop details from the cluster configuration and extracts the object metadata from the Hadoop cluster.
4. The Metadata Access Service returns the metadata to the Developer tool.
5. When you save the data object, the information is saved in the Model repository.

Metadata Access Service Use Cases

You can use the Metadata Access Service in the following scenarios:

Informatica 10.2.1 fresh installation

After the Informatica 10.2.1 installation is complete, you can create a Metadata Access Service. The following table summarizes the dependencies for products, services, and databases that are associated with the Metadata Access Service:

Dependency	Summary
Products	The following products use the Metadata Access Service: - Big Data Management - PowerExchange for HBase - PowerExchange for HDFS - PowerExchange for Hive - PowerExchange for MapR-DB - Big Data Parser - Big Data Quality - Big Data Streaming
Services	The Metadata Access Service is not dependent on other services.
Databases	The Metadata Access Service does not have any associated database.
Installer	You cannot create the Metadata Access Service when you run the installer.

Upgrade to Informatica 10.2.1

After you upgrade to Informatica version 10.2.1, the existing HBase, Hive, HDFS, MapR-DB, and Big Data Streaming mappings that use existing connections will continue to run successfully. The Metadata Access Service accesses the Hadoop environment at design time. If you re-import the source and target objects or change the connection in the existing mappings, ensure that your Informatica administrator creates a Metadata Access Service in the domain.

Create a Metadata Access Service if you want to use HBase, Hive, HDFS, MapR-DB, and Big Data Streaming connections to import an object from a Hadoop cluster. The Metadata Access Service is not dependent on other services. You can create a Metadata Access Service in any order.

Metadata Access Service Usage Guidelines

Use the Metadata Access Service on the Developer tool machine when you import and preview metadata from a Hadoop distribution.

Create a Metadata Access Service before you import HBase, HDFS, Hive, MapR-DB, and MapRStreams objects from a Hadoop distribution. Consider the following guidelines when you import HBase, HDFS, Hive, MapR-DB, and MapRStreams objects:

- If your Informatica administrator creates only one Metadata Access Service in the domain and the same service is set as default, the import process goes seamless.
- If your Informatica administrator creates one or more Metadata Access Services in the domain and none is set as default, you need to select a default Metadata Access Service during the import process or from the Preferences page before starting the import process.
- If your Informatica administrator creates one or more Metadata Access Services in the domain and one service is set as default, the import process goes seamless. However, you can override the service by changing the default Metadata Access Service from the **Preferences** page before starting the import process.

Although a Metadata Access Service is not required to create HBase, HDFS, Hive, MapR-DB, and MapRStreams connections, ensure that a default Metadata Access Service is set before you test HBase, HDFS, Hive, MapR-DB, and MapRStreams connections.

Creating a Metadata Access Service

1. In Informatica Administrator, click the **Manage** tab.
2. Click the **Services and Nodes** view.
3. In the Domain Navigator, select the domain.
4. Click **Actions** > **New** > **Metadata Access Service**.

   The **New Metadata Access Service** wizard appears.
5. On the **New Metadata Access Service - Step 1 of 3** page, enter the following properties:

Property	Description
Name	Name of the service. The name is not case sensitive and must be unique within the domain. It cannot exceed 128 characters or begin with @. It also cannot contain spaces or the following special characters: ` ~ % ^ * + = { } \ ; : ' " / ? . , < > | ! ( )] [
Description	Description of the service. The description cannot exceed 765 characters.
Location	Domain and folder where the service is created. Click **Browse** to choose a different folder. You can move the service after you create it.
License	License object that allows use of the service.
Node	Node on which the service runs.
Backup Nodes	If your license includes high availability, nodes on which the service can run if the primary node is unavailable.

6. Click **Next**.

   The **New Metadata Access Service - Step 2 of 3** page appears.
7. Select the HTTP Protocol Type and enter the respective port number and other security parameters to use for the Metadata Access Service.

Property	Description
HTTP Protocol Type	Security protocol that the Metadata Access Service uses. Select one of the following values: - HTTP. Requests to the service must use an HTTP URL. - HTTPS. Requests to the service must use an HTTPS URL. When you set the HTTP protocol type to HTTPS, you enable Transport Layer Security (TLS) for the service. You must configure the HTTP or HTTPS port for each service process. Default is HTTP.
HTTP Port	HTTP port. Default is 7080.
HTTPS Port	HTTPS port.

Keystore File	Path and file name of the keystore file that contains the keys and certificates required if you use HTTPS connections for the Metadata Access Service. You can create a keystore file with a keytool. keytool is a utility that generates and stores private or public key pairs and associated certificates in a keystore file. You can use the self-signed certificate or use a certificate signed by a certificate authority.
Keystore Password	Password for the keystore file.
Truststore File	Path and file name of the truststore file that contains authentication certificates trusted by the Metadata Access Service.
Truststore Password	Password for the truststore file.
SSL Protocol	Secure Sockets Layer protocol to use. Default is TLS.
Enable Service	Enable the Metadata Access Service when it is created. The Metadata Access Service is not dependant on any other service.

8. Click **Next**.

   The **New Metadata Access Service - Step 3 of 3** page appears.

9. If applicable, specify the execution options for the Kerberos cluster.

Property	Description
Hadoop Kerberos Service Principal Name	The Service Principal Name (SPN) of the Metadata Access Service to connect to a Hadoop cluster that uses Kerberos authentication.
Hadoop Kerberos Keytab	The file path to the Kerberos keytab file on the machine on which the Metadata Access Service runs.
Use logged in user as impersonation user	Required if the Hadoop cluster uses Kerberos authentication. If enabled, the Metadata Access Service uses the current logged in user, in the Developer tool, as the impersonation user while accessing any resources. Default is false.

10. Click **Finish**.

    The domain creates and enables the Metadata Access Service.

Metadata Access Service FAQs

Why do I create a Metadata Access Service?

The Metadata Access Service allows the Developer tool to access Hadoop connection information and to import and preview metadata from a Hadoop distribution.

Can I create multiple Metadata Access Services in a domain?

Yes. If you have multiple Hadoop distributions configured in your domain, Informatica recommends creating a separate Metadata Access Service for each Hadoop distribution.

Can I use a single Metadata Access Service for multiple Hadoop distributions?

Yes. You need to recycle the service to use it for a different Hadoop distribution. However, Informatica does not recommend using a single Metadata Access Service for multiple Hadoop distributions. If you use a common Metadata Access Service instance for different Hadoop distributions, you might face exceptions.

How can I associate a Metadata Access Service with a particular Hadoop distribution?

Set the HadoopDistribution custom property to associate a Metadata Access Service with a particular Hadoop distribution.

1. In Informatica Administrator, click the **Manage** tab > **Services and Nodes** view.
2. In the Domain Navigator, select the Metadata Access Service.
3. Click the **Processes** tab.
4. Click **Custom Properties**. The **Edit Custom Properties** dialog box appears.
5. Click **New** to add the `HadoopDistribution` custom property.
   `ExecutionContextOptions.HadoopDistribution=<distribution>_<version>`

   For example, `ExecutionContextOptions.HadoopDistribution=CDH_5.13`
6. Recycle the Metadata Access Service.

How can I import metadata from multiple Hadoop distributions?

Ensure that your Informatica administrator creates a separate Metadata Access Service instance for each Hadoop distribution. Select a different Metadata Access Service from the Preferences page before starting the import process.

If I upgrade from previous versions of Informatica to Informatica 10.2.1, will the existing HBase, HDFS, Hive, MapR-DB, and MapRStreams mappings continue to work?

The existing mappings without any modification will continue to work. If you re-import the source or target objects, make sure that a Metadata Access Service exists to import and preview metadata successfully.

I am unable to import metadata for a local complex file even if a Metadata Access Service exists in the domain.

When you import a complex file of Avro or Parquet, the imported object includes the metadata associated with the distribution in the Hadoop environment. If the file is local or the file resides on the Developer tool machine, the import process picks up the distribution information from the `developerCore.ini` file. You must edit the `developerCore.ini` file to point to the distribution directory on the Developer tool machine. You can find the `developerCore.ini` file in the following directory:

```
<Informatica installation directory>\clients\DeveloperClient
```

Add the following property:

```
-DINFA_HADOOP_DIST_DIR=hadoop\<distribution>_<version>
```

For example, `-DINFA_HADOOP_DIST_DIR=hadoop\mapr_5.2.0`

The following error appears when I create a Metadata Access Service:

```
[DOM_10079] Cannot start service <metadata_access_service_name> on any node specified for the service.
```

The port number you specified for the Metadata Access Service might be used by another service. Verify and update the port number. Restart the Metadata Access Service.

infacmd mas Command Reference

Use *infacmd* command line options to create Metadata Access Service objects and processes.

The *infacmd* plugin ID for the Metadata Access Service is `mas`. *infacmd* provides the following command line options for the Metadata Access Service:

- `CreateService`
- `ListServiceOptions`
- `ListServiceProcessOptions`
- `UpdateServiceOptions`
- `UpdateServiceProcessOptions`

CreateService

Creates a Metadata Access Service. The Metadata Access Service is an application service that allows the Developer tool to access Hadoop connection information to import and preview metadata.

The infacmd mas CreateService command uses the following syntax:

```
CreateService
<-DomainName|-dn> DomainName
<-NodeName|-nn> NodeName
<-UserName|-un> Username
<-Password|-pd> Password
<-ServiceName|-sn> ServiceName
<-HTTPProtocolType|-hp> HTTPProtocolType
[<-HTTPPort|-pt> HTTPPort]
[<-HTTPSPort|-spt> HTTPSPort]
[<-HadoopServicePrincipalName|-hpn> HadoopServicePrincipalName]
[<-HadoopKeyTab|-hkt> HadoopKeyTab]
[<-ServiceDescription|-sd> ServiceDescription]
[<-ResilienceTimeout|-re> ResilienceTimeout]
[<-FolderPath|-fp> FolderPath]
[<-BackupNodes|-bn> BackupNodes]
[<-KeyStoreFile|-kf> KeyStoreFile]
[<-KeystorePassword|-kp> KeystorePassword]
[<-TruststoreFile|-tf> TruststoreFile]
[<-TruststorePassword|-tp> TruststorePassword]
[<-SecurityDomain|-sdn> SecurityDomain]
[<-SSLProtocol|-sp> SSLProtocol]
[<-HadoopImpersonationUser|-hu> hadoop_impersonation_user]
[<-loggedInUserAsImpersonationUser|-uiu> UseLoggedInUserAsImpersonationUser]
```

The following table describes infacmd mas CreateService options and arguments:

Option	Argument	Description
-DomainName -dn	domain_name	Required. Name of the Informatica domain. You can set the domain name with the -dn option or the environment variable INFA_DEFAULT_DOMAIN. If you set a domain name with both methods, the -dn option takes precedence.
-NodeName -nn	node_name	Node where the Metadata Access Service runs. You can run the Data Integration Service only on a node.
-UserName -un	user_name	Required if the domain uses Native or LDAP authentication. User name to connect to the domain. You can set the user name with the -un option or the environment variable INFA_DEFAULT_DOMAIN_USER. If you set a user name with both methods, the -un option takes precedence. Optional if the domain uses Kerberos authentication. To run the command with single sign-on, do not set the user name. If you set the user name, the command runs without single sign-on.
-Password -pd	password	Required if you specify the user name. Password for the user name. The password is case sensitive. You can set a password with the -pd option or the environment variable INFA_DEFAULT_DOMAIN_PASSWORD. If you set a password with both methods, the password set with the -pd option takes precedence.
-ServiceName -sn	service_name	Required. Name of the Metadata Access Service. The name is not case sensitive and must be unique within the domain. The characters must be compatible with the code page of the associated repository. The name cannot exceed 230 characters, have leading or trailing spaces, or contain carriage returns, tabs, or the following characters: / * ? < > " |
-HTTPProtocolType -hp	http_protocol_type	Security protocol that the Metadata Access Service uses. Enter one of the following values: - HTTP. Requests to the service must use an HTTP URL. - HTTPS. Requests to the service must use an HTTPS URL. When you set the HTTP protocol type to HTTPS, you enable Transport Layer Security (TLS) for the service. Default is HTTP.
-HTTPPort -pt	http_port	Required if you do not specify an HTTPS port. Unique HTTP port number used for each Metadata Access Service process. After you create the service, you can define different port numbers for each Metadata Access Service process. Default is 7080.
-HTTPSPort -spt	https_port	Required if you do not specify an HTTP port. Unique HTTPS port number used for each Metadata Access Service process. After you create the service, you can define different port numbers for each Metadata Access Service process.

Option	Argument	Description
-HadoopServicePrincipalName -hpn	hadoop_spn	Service Principal Name (SPN) of the Metadata Access Service to connect to a Hadoop cluster that uses Kerberos authentication.
-HadoopKeyTab -hkt	keytab_file_path	The file path to the Kerberos keytab file on the machine on which the Metadata Access Service runs.
-ServiceDescription -sd	service_description	Optional. Service description.
-ResilienceTimeout -re	timeout_period_in_seconds	Optional. Amount of time in seconds that infacmd attempts to establish or re-establish a connection to the domain. You can set the resilience timeout period with the -re option or the environment variable INFA_CLIENT_RESILIENCE_TIMEOUT. If you set the resilience timeout period with both methods, the -re option takes precedence.
-FolderPath -fp	full_folder_path	Optional. Full path, excluding the domain name, to the folder in which you want to create the Metadata Access Service. Must be in the following format: `/parent_folder/child_folder` Default is "/" (the domain).
-BackupNodes -bn	node_name1,node_name2,...	Optional. Nodes on which the service can run if the primary node is unavailable. You can configure backup nodes if you have high availability.
-KeystoreFile -kf	keystore_file_location	Path and file name of the keystore file that contains the keys and certificates required if you use the HTTPS protocol for the Metadata Access Service. You can create a keystore file with a keytool. keytool is a utility that generates and stores private or public key pairs and associated certificates in a keystore file. You can use the self-signed certificate or use a certificate signed by a certificate authority.
-KeystorePassword -kp	keystore_password	Password for the keystore file.
-TruststoreFile -tf	trust_store_file	Required when the domain is SSL- enabled. Domain truststore file location in the cluster.
-TruststorePassword -tp	trust_store_password	Required when the domain is SSL- enabled. Truststore domain password.

Option	Argument	Description
-SecurityDomain -sdn	security_domain	Required if the domain uses LDAP authentication. Optional if the domain uses native authentication or Kerberos authentication. Name of the security domain to which the domain user belongs. You can set a security domain with the -sdn option or the environment variable INFA_DEFAULT_SECURITY_DOMAIN. If you set a security domain name with both methods, the -sdn option takes precedence. The security domain name is case sensitive. If the domain uses native or LDAP authentication, the default is Native. If the domain uses Kerberos authentication, the default is the LDAP security domain created during installation. The name of the security domain is the same as the user realm specified during installation.
-SSLProtocol -sp	ssl_protocol	Optional. Secure Sockets Layer protocol to use.
-HadoopImpersonationUser|- -hu	hadoop_impersonation_user	Optional. Enter a user name for the Metadata Access Service to impersonate when it connects to a Hadoop environment.
- loggedInUserAsImpersonationUser -uiu	use_logged_in_user_as_proxy	Required if the Hadoop cluster uses Kerberos authentication. Hadoop impersonation user. The user name that the Metadata Access Service impersonates to import metadata from the Hadoop environment at design time.

ListServiceOptions

Lists the properties for a Metadata Access Service.

The infacmd mas ListServiceOptions command uses the following syntax:

```
ListServiceOptions
<-DomainName|-dn> DomainName
<-UserName|-un> Username
<-Password|-pd> Password
<-ServiceName|-sn> ServiceName
[<-SecurityDomain|-sdn> SecurityDomain]
[<-ResilienceTimeout|-re> ResilienceTimeout]
```

The following table describes infacmd mas ListServiceOptions options and arguments:

Option	Argument	Description
-DomainName -dn	domain_name	Required. Name of the Informatica domain. You can set the domain name with the -dn option or the environment variable INFA_DEFAULT_DOMAIN. If you set a domain name with both methods, the -dn option takes precedence.
-UserName -un	user_name	Required if the domain uses Native or LDAP authentication. User name to connect to the domain. You can set the user name with the -un option or the environment variable INFA_DEFAULT_DOMAIN_USER. If you set a user name with both methods, the -un option takes precedence. Optional if the domain uses Kerberos authentication. To run the command with single sign-on, do not set the user name. If you set the user name, the command runs without single sign-on.
-Password -pd	password	Required if you specify the user name. Password for the user name. The password is case sensitive. You can set a password with the -pd option or the environment variable INFA_DEFAULT_DOMAIN_PASSWORD. If you set a password with both methods, the password set with the -pd option takes precedence.
-ServiceName -sn	service_name	Required. Name of the Metadata Access Service.
-SecurityDomain -sdn	security_domain	Required if the domain uses LDAP authentication. Optional if the domain uses native authentication or Kerberos authentication. Name of the security domain to which the domain user belongs. You can set a security domain with the -sdn option or the environment variable INFA_DEFAULT_SECURITY_DOMAIN. If you set a security domain name with both methods, the -sdn option takes precedence. The security domain name is case sensitive. If the domain uses native or LDAP authentication, the default is Native. If the domain uses Kerberos authentication, the default is the LDAP security domain created during installation. The name of the security domain is the same as the user realm specified during installation.
-ResilienceTimeout -re	timeout_period_in_seconds	Optional. Amount of time in seconds that infacmd attempts to establish or re-establish a connection to the domain. You can set the resilience timeout period with the -re option or the environment variable INFA_CLIENT_RESILIENCE_TIMEOUT. If you set the resilience timeout period with both methods, the -re option takes precedence.

ListServiceProcessOptions

Lists the properties of a Metadata Access Service process.

The infacmd mas ListServiceProcessOptions command uses the following syntax:

```
ListServiceProcessOptions
```

```
<-DomainName|-dn> DomainName

<-NodeName|-nn> NodeName

<-UserName|-un> Username

<-Password|-pd> Password

<-ServiceName|-sn> ServiceName

[<-SecurityDomain|-sdn> SecurityDomain]

[<-ResilienceTimeout|-re> ResilienceTimeout]
```

The following table describes infacmd mas ListServiceProcessOptions options and arguments:

Option	Argument	Description
-DomainName -dn	domain_name	Required. Name of the Informatica domain. You can set the domain name with the -dn option or the environment variable INFA_DEFAULT_DOMAIN. If you set a domain name with both methods, the -dn option takes precedence.
-NodeName -nn	node_name	Required. Name of node where the service process runs.
-UserName -un	user_name	Required if the domain uses Native or LDAP authentication. User name to connect to the domain. You can set the user name with the -un option or the environment variable INFA_DEFAULT_DOMAIN_USER. If you set a user name with both methods, the -un option takes precedence. Optional if the domain uses Kerberos authentication. To run the command with single sign-on, do not set the user name. If you set the user name, the command runs without single sign-on.
-Password -pd	password	Required if you specify the user name. Password for the user name. The password is case sensitive. You can set a password with the -pd option or the environment variable INFA_DEFAULT_DOMAIN_PASSWORD. If you set a password with both methods, the password set with the -pd option takes precedence.
-ServiceName -sn	service_name	Required. Name of the Metadata Access Service.

Option	Argument	Description
-SecurityDomain -sdn	security_domain	Required if the domain uses LDAP authentication. Optional if the domain uses native authentication or Kerberos authentication. Name of the security domain to which the domain user belongs. You can set a security domain with the -sdn option or the environment variable INFA_DEFAULT_SECURITY_DOMAIN. If you set a security domain name with both methods, the -sdn option takes precedence. The security domain name is case sensitive. If the domain uses native or LDAP authentication, the default is Native. If the domain uses Kerberos authentication, the default is the LDAP security domain created during installation. The name of the security domain is the same as the user realm specified during installation.
-ResilienceTimeout -re	timeout_period_in_seconds	Optional. Amount of time in seconds that infacmd attempts to establish or re-establish a connection to the domain. You can set the resilience timeout period with the -re option or the environment variable INFA_CLIENT_RESILIENCE_TIMEOUT. If you set the resilience timeout period with both methods, the -re option takes precedence.

UpdateServiceOptions

Updates Metadata Access Service properties. To view current properties run the infacmd mas ListServiceOptions command.

You can change the properties while the service is running, but you must recycle the service for the changed properties to take effect.

The infacmd mas UpdateServiceOptions command uses the following syntax:

```
UpdateServiceOptions
<-DomainName|-dn> DomainName
<-UserName|-un> Username
<-Password|-pd> Password
<-ServiceName|-sn> ServiceName
[<-SecurityDomain|-sdn> SecurityDomain]
[<-ResilienceTimeout|-re> ResilienceTimeout]
[<-Options|-o> options]
<-PrimaryNode|-nn> PrimaryNodeName
[<-BackupNodes|-bn> node_name1,node_name2,...]
[<-SearchIndexRoot|-si> SearchIndexRoot]
```

The following table describes infacmd mas UpdateServiceOptions options and arguments:

Option	Argument	Description
-DomainName -dn	domain_name	Required. Name of the Informatica domain. You can set the domain name with the -dn option or the environment variable INFA_DEFAULT_DOMAIN. If you set a domain name with both methods, the -dn option takes precedence.
-UserName -un	user_name	Required if the domain uses Native or LDAP authentication. User name to connect to the domain. You can set the user name with the -un option or the environment variable INFA_DEFAULT_DOMAIN_USER. If you set a user name with both methods, the -un option takes precedence. Optional if the domain uses Kerberos authentication. To run the command with single sign-on, do not set the user name. If you set the user name, the command runs without single sign-on.
-Password -pd	password	Required if you specify the user name. Password for the user name. The password is case sensitive. You can set a password with the -pd option or the environment variable INFA_DEFAULT_DOMAIN_PASSWORD. If you set a password with both methods, the password set with the -pd option takes precedence.
-ServiceName -sn	service_name	Required. Name of the Metadata Access Service that the application is deployed to.
-SecurityDomain -sdn	security_domain	Required if the domain uses LDAP authentication. Optional if the domain uses native authentication or Kerberos authentication. Name of the security domain to which the domain user belongs. You can set a security domain with the -sdn option or the environment variable INFA_DEFAULT_SECURITY_DOMAIN. If you set a security domain name with both methods, the -sdn option takes precedence. The security domain name is case sensitive. If the domain uses native or LDAP authentication, the default is Native. If the domain uses Kerberos authentication, the default is the LDAP security domain created during installation. The name of the security domain is the same as the user realm specified during installation.
-ResilienceTimeout -re	timeout_period_in_seconds	Optional. Amount of time in seconds that infacmd attempts to establish or re-establish a connection to the domain. You can set the resilience timeout period with the -re option or the environment variable INFA_CLIENT_RESILIENCE_TIMEOUT. If you set the resilience timeout period with both methods, the -re option takes precedence.
-Options -o	options	Optional. Enter each option separated by a space. To view options, run the infacmd mas ListServiceOptions command.
-PrimaryNode -nn	node_name	Enter the node where the Metadata Access Service will run. The Metadata Access Service can run only on a node.

Option	Argument	Description
-BackupNodes -bn	node_name1,node_name2,...	Optional. Nodes on which the service can run if the primary node is unavailable. You can configure backup nodes if you have high availability.
-SearchIndexRoot -si	search_index_root	Optional. Changes the directory for the search index. Enter the full path to the directory. Default is the Informatica installation directory.

Metadata Access Service Options

Use the Metadata Access Service options with the infacmd mas UpdateServiceOptions command.

Enter Metadata Access Service options in the following format:

```
... -o option_type.option_name=value
```

To enter multiple options, separate them with a space. To enter a value that contains a space or other non-alphanumeric character, enclose the value in quotation marks.

The following table describes Metadata Access Service options:

Option	Description
ExecutionContextOptions.HadoopDistribution	The Hadoop distribution directory on the Metadata Access Service node. The contents of the Metadata Access Service Hadoop distribution directory must be identical to Hadoop distribution directory on the data nodes. Type `<Informatica Installation directory/Informatica/services/shared/hadoop/[Hadoop_distribution_name].`
HttpConfigurationOptions.HTTPProtocolType	Security protocol that the Metadata Access Service uses. Enter one of the following values: - HTTP. Requests to the service must use an HTTP URL. - HTTPS. Requests to the service must use an HTTPS URL. When you set the HTTP protocol type to HTTPS, you enable Transport Layer Security (TLS) for the service. Default is HTTP.
MASProperties.HadoopKeytab	The file path to the Kerberos keytab file on the machine on which the Metadata Access Service runs.
MASProperties.HadoopPrincipal	Service Principal Name (SPN) of the Metadata Access Service to connect to a Hadoop cluster that uses Kerberos authentication.
MASProperties.LoggedInUserAsImperUser	Required if the Hadoop cluster uses Kerberos authentication. Hadoop impersonation user. The user name that the Metadata Access Service impersonates to run mappings in the Hadoop environment.

UpdateServiceProcessOptions

Updates properties for a Metadata Access Service process. To view current properties, run the infacmd mas ListServiceProcessOptions command.

Enter options in the following format:

```
... -o option_name=value option_name=value ...
```

Separate multiple options with a space. To enter a value that contains a space or other non-alphanumeric character, enclose the value in quotation marks.

The infacmd mas UpdateServiceProcessOptions command uses the following syntax:

```
UpdateServiceProcessOptions
<-DomainName|-dn> DomainName
<-NodeName|-nn> NodeName
<-UserName|-un> Username
<-Password|-pd> Password
<-ServiceName|-sn> ServiceName
[<-SecurityDomain|-sdn> SecurityDomain]
[<-ResilienceTimeout|-re> ResilienceTimeout]
```

The following table describes infacmd mas UpdateServiceProcessOptions options and arguments:

Option	Argument	Description
-DomainName -dn	domain_name	Required. Name of the Informatica domain. You can set the domain name with the -dn option or the environment variable INFA_DEFAULT_DOMAIN. If you set a domain name with both methods, the -dn option takes precedence.
NodeName -nn	node_name	Required. Node where the Metadata Access Service runs.
-UserName -un	user_name	Required if the domain uses Native or LDAP authentication. User name to connect to the domain. You can set the user name with the -un option or the environment variable INFA_DEFAULT_DOMAIN_USER. If you set a user name with both methods, the -un option takes precedence. Optional if the domain uses Kerberos authentication. To run the command with single sign-on, do not set the user name. If you set the user name, the command runs without single sign-on.
-Password -pd	password	Required if you specify the user name. Password for the user name. The password is case sensitive. You can set a password with the -pd option or the environment variable INFA_DEFAULT_DOMAIN_PASSWORD. If you set a password with both methods, the password set with the -pd option takes precedence.
-ServiceName -sn	service_name	Required. Name of the Metadata Access Service.

Option	Argument	Description
-SecurityDomain -sdn	security_domain	Required if the domain uses LDAP authentication. Optional if the domain uses native authentication or Kerberos authentication. Name of the security domain to which the domain user belongs. You can set a security domain with the -sdn option or the environment variable INFA_DEFAULT_SECURITY_DOMAIN. If you set a security domain name with both methods, the -sdn option takes precedence. The security domain name is case sensitive. If the domain uses native or LDAP authentication, the default is Native. If the domain uses Kerberos authentication, the default is the LDAP security domain created during installation. The name of the security domain is the same as the user realm specified during installation.
-ResilienceTimeout -re	timeout_period_in_seconds	Optional. Amount of time in seconds that infacmd attempts to establish or re-establish a connection to the domain. You can set the resilience timeout period with the -re option or the environment variable INFA_CLIENT_RESILIENCE_TIMEOUT. If you set the resilience timeout period with both methods, the -re option takes precedence.

Metadata Access Service Process Options

Use the Metadata Access Service process options with the infacmd mas UpdateServiceProcessOptions command.

Enter Metadata Access Service process options in the following format:

```
... -o option_type.option_name=value
```

To enter multiple options, separate them with a space. To enter a value that contains a space or other non-alphanumeric character, enclose the value in quotation marks.

The following table describes Metadata Access Service process options:

Option	Description
GeneralOptions.JVMOptions	Java Virtual Machine (JVM) command line options to run Java-based programs. When you configure the JVM options, you must set the Java SDK classpath, Java SDK minimum memory, and Java SDK maximum memory properties.
HttpConfigurationOptions.KeyStoreFile	Path and file name of the keystore file that contains the keys and certificates required if you use the HTTPS protocol for the Metadata Access Service. You can create a keystore file with a keytool. keytool is a utility that generates and stores private or public key pairs and associated certificates in a keystore file. You can use the self-signed certificate or use a certificate signed by a certificate authority.
HttpConfigurationOptions.KeyStorePassword	Password for the keystore file.
HttpConfigurationOptions.MaxBacklogRequests	Maximum number of HTTP or HTTPS connections that can wait in a queue for this Metadata Access Service process. Default is 100.

Option	Description
HttpConfigurationOptions.MaxConcurrentRequests	Maximum number of HTTP or HTTPS connections that can be made to this Metadata Access Service process. Minimum is 4. Default is 200.
HttpConfigurationOptions.SSLProtocol	Secure Sockets Layer protocol to use. Default is TLS.
HttpConfigurationOptions.TrustStoreFile	Path and file name of the truststore file that contains authentication certificates trusted by the Metadata Access Service.
HttpConfigurationOptions.TrustStorePassword	Password for the truststore file.

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